

**THE ELEVENTH WILLIAM LOWELL PUTNAM
MATHEMATICAL COMPETITION**

March 31, 1951

Morning Session

1. Show that the determinant:

$$\begin{vmatrix} 0 & a & b & c \\ -a & 0 & d & e \\ -b & -d & 0 & f \\ -c & -e & -f & 0 \end{vmatrix}$$

is non-negative, if its elements a, b, c , etc., are real. (page 317)

2. In the plane, what is the locus of points the sum of the squares of whose distances from n fixed points is a constant? What restrictions, stated in geometric terms, must be put on the constant so that the locus is non-null? (page 319)

3. Find the sum to infinity of the series:

$$1 - \frac{1}{4} + \frac{1}{7} - \frac{1}{10} + \dots + \frac{(-1)^{n+1}}{3n-2} + \dots \quad (\text{page 319})$$

4. Trace the curve whose equation is:

$$y^4 - x^4 - 96y^2 + 100x^2 = 0. \quad (\text{page 320})$$

5. Consider in the plane the network of points having integral coordinates. For lines having rational slope show that:

(i) the line passes through no points of the network or through infinitely many;

(ii) there exists for each line a positive number d having the property that no point of the network, except such as may be on the line, is closer to the line than the distance d . (page 323)

6. Determine the position of a normal chord of a parabola such that it cuts off of the parabola a segment of minimum area. (page 324)

7. Show that if the series $a_1 + a_2 + a_3 + \cdots + a_n + \cdots$ converges, then the series $a_1 + a_2/2 + a_3/3 + \cdots + a_n/n + \cdots$ converges also.

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Afternoon Session

1. Find the condition that the functions $M(x, y)$ and $N(x, y)$ must satisfy in order that the differential equation $Mdx + Ndy = 0$ shall have an integrating factor of the form $f(xy)$. You may assume that M and N have continuous partial derivatives of all orders.

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2. Two functions of x are differentiable and not identically zero. Find an example of two such functions having the property that the derivative of their quotient is the quotient of their derivatives.

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3. Show that if x is positive, then

$$\log_e(1 + 1/x) > 1/(1 + x). \quad (\text{page 329})$$

4. Investigate, in any way which yields significant results, the existence, in the plane, of the configuration consisting of an ellipse simultaneously tangent to four distinct concentric circles.

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5. A plane through the center of a torus is tangent to the torus. Prove that the intersection of the plane and the torus consists of two circles.

(page 334)

6. Assuming that all the roots of the cubic equation $x^3 + ax^2 + bx + c = 0$ are real, show that the difference between the greatest and the least roots is not less than $(a^2 - 3b)^{1/2}$ or greater than $2(a^2 - 3b)^{1/2}/3^{1/2}$.

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7. Find the volume of the four-dimensional hypersphere $x^2 + y^2 + z^2 + t^2 = r^2$, and also the hypervolume of its interior $x^2 + y^2 + z^2 + t^2 < r^2$.

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